

Origins of Plasma during spallation drilling

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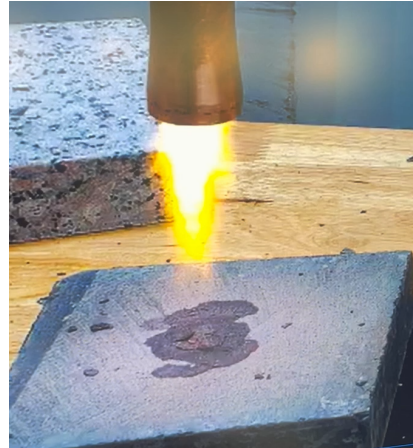
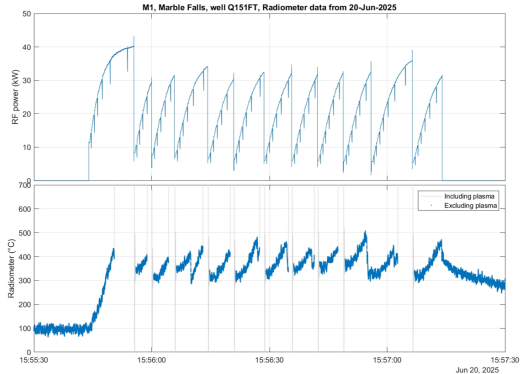
QUAISE

September 1, 2026

Overview

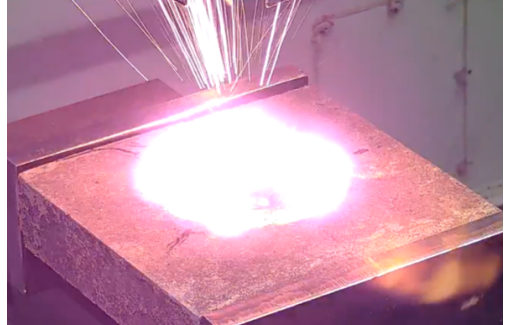
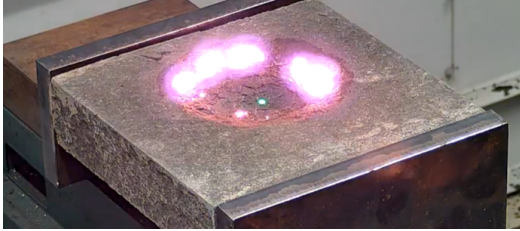
- 1 Observations
- 2 Plasma physics
- 3 Simulation
- 4 Results

Evidence of plasma breakdown events



$\omega < \omega_p$ - gas breakdown (shielding)

Evidence of thermionic emission (cold plasma $T \leq 2700$ K)



$\omega > \omega_p$ - surface ionisation (no shielding)

What are the physical phenomena involved?

- 1 Saha gas ionisation: state of ionisation of a gas (or electron density) under thermal equilibrium.

- $K \rightleftharpoons K^+ + e^-$ with $E_I^K = 4.34$ eV
- Contribution above $T > 3000$ K (beam absorption)

- 2 Surface ionisation (thermionic effect): n_e at the surface of a spall under thermal equilibrium.

- $\frac{n_e \cdot n_{K^+}}{n_K} = A(T) \exp\left(-\frac{W_f^g \cdot Q_E}{k_B \cdot T}\right)$ with $W_f \approx 2.5$ eV
- Contribution above $T > 2700$ K (beam absorption)
- For $\lambda = 3$ mm, coarse spalls ($100 - 3000 \mu\text{m}$) are efficient absorbers $\rightarrow$ seed the plasma.

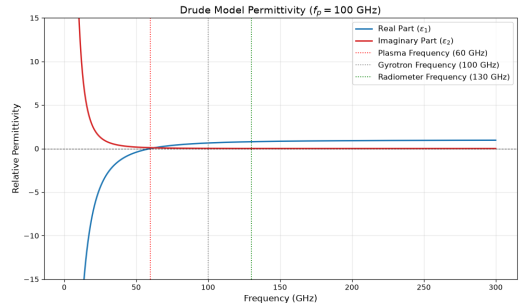


Figure: $f = 100$ GHz, $f_p = 60$ GHz, $\nu_c = 10$ GHz

Air breakdown (electron avalanche)

- ❶ Breakdown occurs when E_{field} of a focused microwave beam strips electrons from air molecules, turning the gas into a highly conductive, glowing plasma
- ❷ Avalanche Ionization:
 - Once "seed" electrons exist $\rightarrow$ absorb energy directly from the E_{field} through a process called inverse Bremsstrahlung (free-free absorption) (negligible in our case)
 - An electron gains kinetic energy through successive wave cycles and random elastic collisions with neutral molecules
 - $KE^e > E_I^g \Rightarrow e^- + M \rightarrow M^+ + 2e^-$
 - exponential growth of electron density $n_e \propto e^{\alpha t}$
 - very fast timescale (ns $\rightarrow$ no control unless ps pulses)
- ❸ $E_{breakdown} \propto E_0 \cdot \frac{P \cdot T_0}{P_0 \cdot T} \cdot \sqrt{1 + \left(\frac{\omega}{\nu_e(P, T)}\right)^2} \cdot f_{reduction} \approx 3 \text{ MV/m}$
- ❹ $\frac{E_{breakdown}}{E_{field}} \approx 70 - 100 \rightarrow \text{no breakdown!}$

Field enhancement due to irregular particles

- 1 Oblate (paraboloid) particle with tip curvature ρ
- 2 E_{field} solution is known analytically $\rightarrow E_{\text{tip}} = E_0 \cdot \frac{a}{\rho} + E_0 \times \text{correction terms}$
- 3 $\Rightarrow$ enhancement factor $\beta_{\text{tip}} = \frac{E_{\text{tip}}}{E_0} \approx \frac{a}{\rho}$
- 4 $a = 1.5 \text{ mm}, 10\mu\text{m} \leq \rho \leq 100\mu\text{m} \Rightarrow 15 \leq \beta_{\text{tip}} \leq 150$
- 5 Avalanche can take place at the tip of spalls
- 6 $E_{\text{breakdown}}$ is reduced by the presence of K in the spall $\Rightarrow$ avalanche at lower β_{tip} , but would need calibration
- 7 Main mechanism given experimental observations

Millimetre wave spallation model with plasma effects

- 1 **Dielectric map:** $n_{\text{eff}}(n_{\text{rock}}, n_{\text{fluid}}, T, \phi, \phi_{\text{crack}}, s_w)$
- 2 **Thermionic effect:** for ejecta absorbed power $\propto d^3$ while cooling $\propto d^2 \rightarrow$ mm spalls $\approx 2000 - 2400$ K for $T_{\text{rock}} = 700$ K $\rightarrow$ seed electrons $\rightarrow$ Drude permittivity.
- 3 **Avalanche test:** compute margin $= \beta \cdot \frac{E_{\text{beam}}}{E_{bd}(p, T, \omega)}$. Ignition when margin ≥ 1 & coarse ejecta present in the beam (≈ 1.5 mm) & minimum volume of ejecta.
- 4 **Bidirectional beam propagation:** (cylindrical coordinates)
- 5 **Absorbed power:** split between rock & plasma: power absorbed by the plasma is a shielding loss that never reaches the rock face
- 6 **Beam gating:** $u(t) = u_{\text{control}} \cdot u_{\text{avoid}} \cdot g_{\text{interlock}}$ with $u_{\text{avoid}} = \left(\frac{\text{margin}_{\text{target}}}{\text{margin}}\right)^2$
- 7 **pore pressure:** $p(T)$
- 8 **Principal effective stresses**
- 9 **Damage model**
- 10 **Detachment & spall removal**

Base simulation (No plasma trip) at 2 bar in air

Thermal spallation: granite, flattop, 2.5 kW

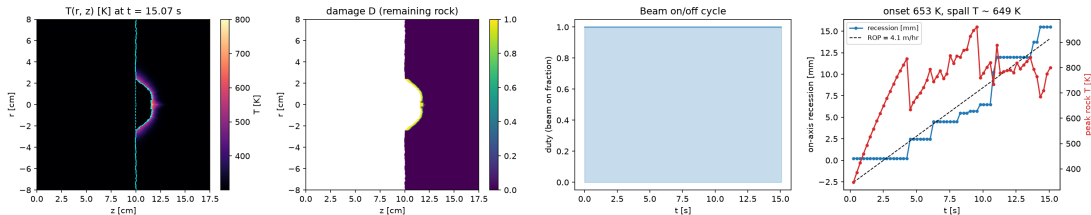
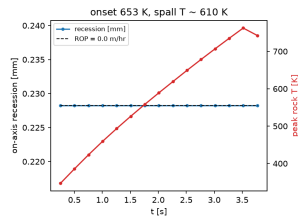
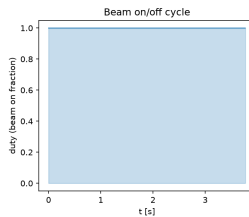
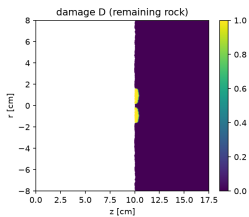
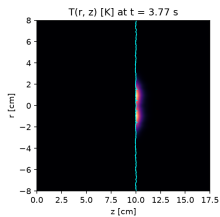


Figure: ROP=4.1 m/hr, TSE = 2.2 kJ/cc

Simulation with Plasma trip (t=3.5s) at 2 bar in air

Thermal spallation: granite, flattop, 2.5 kW



Ramp and switch on/of control $t_{off}=1s$ at 2 bar in air

Thermal spallation: granite, flattop, 3.5 kW

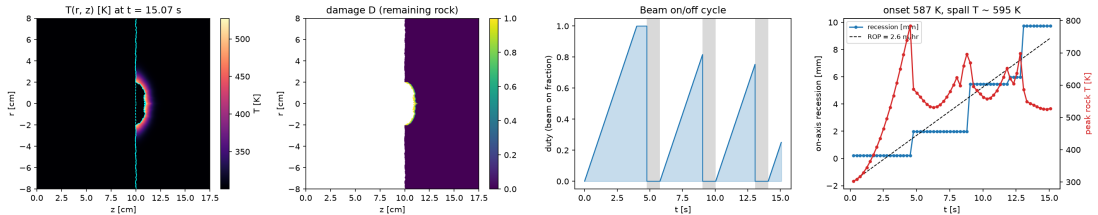


Figure: ROP = 2.6 m/hr, TSE=2.2 kJ/cc

Switch on/off control ($t_{off}=50\text{ms}$) at 2 bar in air

Thermal spallation: granite, flattop, 2.5 kW

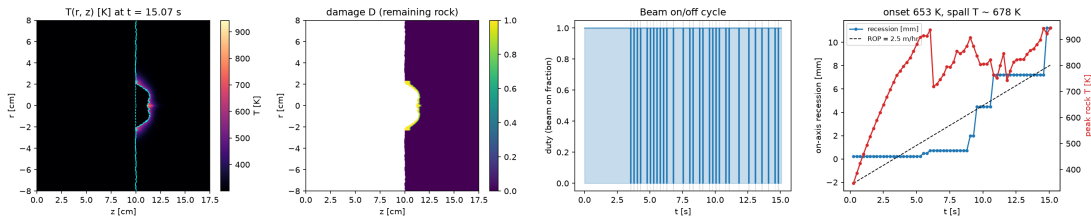
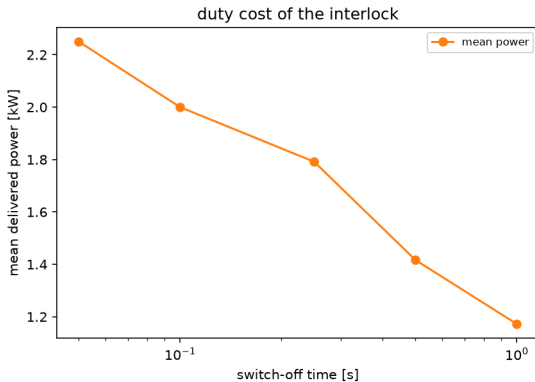
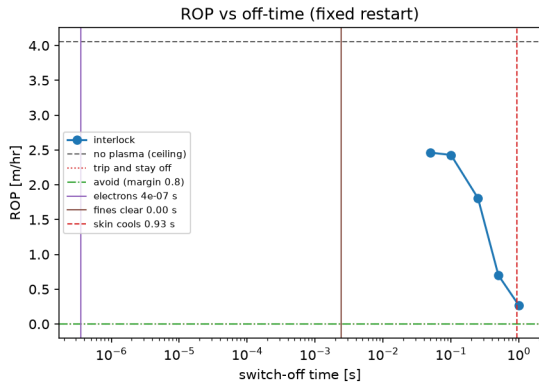


Figure: ROP = 2.5 m/hr , TSE = 2.7 kJ/cc

Summary of switch on/off study at 2 bar



Ramp and switch on/of control $t_{off}=1s$ at 15 bar in air

Thermal spallation: granite, flattop, 3.5 kW

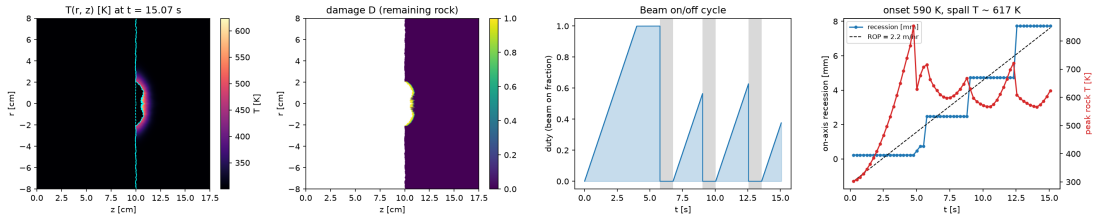


Figure: ROP = 2.2 m/hr, TSE=2.82 kJ/cc

Switch on/off control ($t_{off}=50\text{ms}$) at 15 bar in air

Thermal spallation: granite, flattop, 2.5 kW

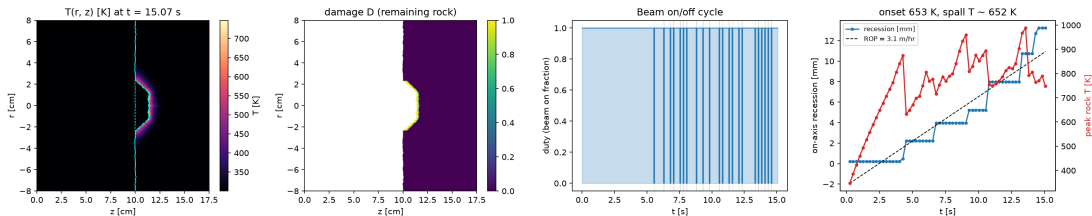
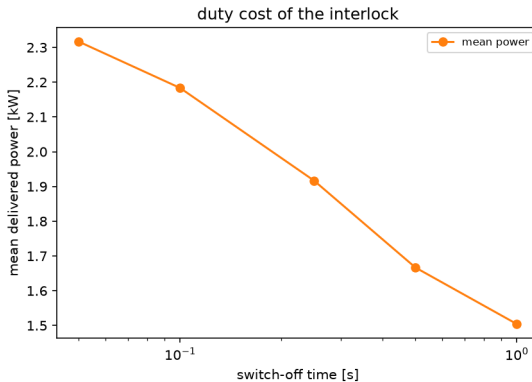
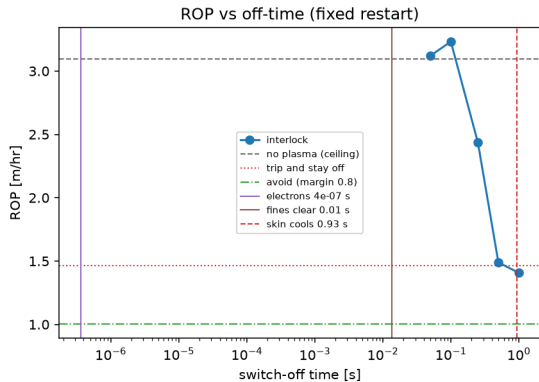


Figure: ROP 3.1 m/hr, TSE = 2 KJ/cc

Summary of switch on/off study at 15 bar



Recommendations and validation experiments

- ➊ Design a system to work at $P \geq 50$ bar (Wellbore stability criterion)
- ➋ Measure the collisional induced absorption in Nitrogen and Argon
- ➌ Measure the breakdown field versus pressure in air, Nitrogen and Argon
- ➍ Measure the breakdown field versus pressure in a dusty fluidized gas
- ➎ Perform spallation experiments with Nitrogen and Argon versus input power and measure tripping time
- ➏ Perform spallation experiments with Nitrogen and Argon versus pressure and measure tripping time
- ➐ Perform dust clearance experiments. Investigate eccentric launcher design (better dust clearance)

Combine BPM model with dusty gas model to improve understanding of dusty plasma (Cassandra)

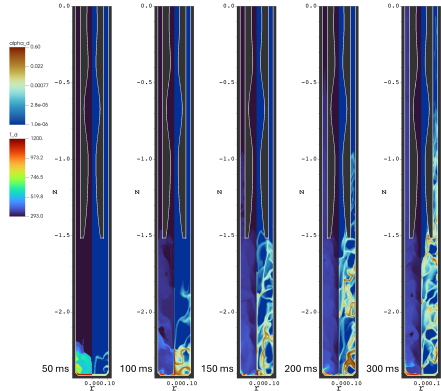


Figure: Dust temperature T_d (left) and dust volume fraction $\log(\alpha_d)$ (right)